Manik Bansal

DRAFT — Industry resume for author review

Computational mechanics | Biomechanics | Scientific computing

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Profile

Computational mechanics researcher with experience in multiscale finite element modeling, fracture mechanics, ocular biomechanics, and blast-induced brain injury modeling. Target role and tailored achievements: pending author input.

Technical skills

MATLAB, Python, Tcl, shell scripting; Linux HPC, SLURM, PSC Bridges-2; LS-DYNA, Abaqus, ANSYS, FEBio; XFEM, explicit/implicit dynamics, multiscale modeling; ParaView, 3D Slicer, ImageJ.

Selected experience

Postdoctoral Researcher — Robert Morris University
Nov 2022–Present (as reported in source CV). High-fidelity human head modeling and multiscale simulations for blast-induced traumatic brain injury research.

Postdoctoral Associate — University of Pittsburgh
Feb 2020–Oct 2022. Micromechanics-based numerical models of scleral collagen architecture and ocular biomechanics.

Senior Research Fellow — IIT Roorkee
Nov 2014–Sep 2019. Computational XFEM and stochastic multiscale frameworks for heterogeneous materials.

Assistant Project Engineer — Sakshi Innovation Pvt. Ltd.
Aug 2009–Nov 2010. Designed and supervised fabrication of special-purpose machine tools for a lightweight expanded clay aggregate plant.

Education

PhD, Mechanical Engineering (Design), IIT Roorkee, 2019.
M.Tech, CAD/CAM, Vellore Institute of Technology, 2013.
B.E., Mechanical Engineering, University of Pune, 2009.

Source: supplied CV dated May 20, 2026. Draft summary; ongoing role and all content require author approval.